PROVED (LEAN 4)

A Formally Verified Divisibility Theorem for A211420

Proof generated by OpenAI GPT-5.6 Pro
Primary Lean formalization by OpenAI Codex
Checked by Lean's kernel

Abstract

Let

$$a_n = \frac{(8n)!\,n!}{(4n)!(3n)!(2n)!}, \qquad L_r = \operatorname{lcm}(1, 2, \ldots, 8r-1).$$

Peter Bala recorded in OEIS A211420 the conjecture that, for each $r \geq 1$ and $k \in \{1, 2, 3\}$, there is a positive integer $C(k, r)$, independent of n, such that

$$\prod_{i=1}^{r} (kn + i) \mid C(k, r) a_n$$

for every $n \geq 0$. We prove the explicit uniform choice

$$C(k, r) = L_r^r.$$

The bound is not claimed to be sharp. The weaker existence assertion follows from classical multivariate Landau theory; the contribution here is the explicit L_r^r bound, a direct prime-power proof, and a complete Lean 4/mathlib formalization checked by Lean's kernel.

Explicit and uniform in n; not claimed sharp.
Historical priority and human peer review are not claimed.

github.com/DannyExperiments/a211420-fixed-divisor